A multi-agent system for RNA design with capability-gated prediction and constraint-navigated generation

Abstract

Computational RNA design still requires experienced biologists to chain many specialized tools. Language model agents promise to automate this process, but face three issues: adopting predictor scores as optimization objectives without validating them on the design task, asserting claims beyond what their evidence supports and reporting gains confounded by unequal sampling budgets. Here we present RNADesignAgent, a multi-agent system for RNA design that addresses these issues through capability-layer evidence gating and constraint navigation. Capability gating evaluates every predictor on verified anchor sets before use and prevents unsupported actions and claims. Constraint navigation guides generators to produce candidates satisfying multi-constraint requirements. Capability gating removed false-positive claims across 45 aptamer tasks (0% versus 54.5% ungated) while preserving most true ones, and constraint navigation improved constraint satisfaction at equal budget across 25 messenger RNA targets (exact McNemar P = 0.009). RNADesignAgent produces high-quality RNA molecules and offers design principles for future scientific agents.

Introduction

RNA therapeutics deliver RNA molecules that encode therapeutic proteins, silence or edit disease-causing genes, or bind disease-causing proteins[1]. Messenger RNAs (mRNAs), small interfering RNAs (siRNAs), guide RNAs (sgRNAs), antisense oligonucleotides (ASOs) and aptamers are approved medicines or late-stage clinical candidates across metabolic, cardiovascular and infectious diseases[2–6]. Designing an RNA molecule with the desired properties is the critical first step in developing these therapies[7]. Experienced computational biologists can accelerate this step and reduce experimental cost by chaining multiple computational tools[8]. However, as the number of computational tools grows and their updates accelerate, even these specialists struggle to master the latest RNA design tools in time[9]. This mismatch between the rapid growth of RNA design tools and the capacity of experts to master them has become a bottleneck in the high-quality computational design of RNA molecules.

Large language model (LLM) agents are promising candidates for this automation, and recent systems have demonstrated agentic tool use in chemistry, protein design, gene-editing experiment planning and biomedical data analysis, but none of these systems covers RNA design[10–15]. Reinforcement learning agents have also been used to steer generative molecular design, including fragment-based 3D generation for protein pockets and diffusion models for structure-based drug design[16,17]. In addition, current scientific agents call tools without validation and report their own results in natural language, and our experiments show that this architecture produces three systematic failure modes. Uncalibrated tool trust occurs when an agent uses a tool's predictive scores without validating whether those predictions have demonstrated power on the specific task. Claim overreach occurs because agents write reports, which often contain statements that exceed what the underlying tools support. Confounded improvement occurs when an agent simply generates more candidates, so that an apparent gain may reflect a larger sampling budget rather than real improvement.

To address these failure modes, we present RNADesignAgent, a multi-agent system for automating RNA design (Fig. 1). The system integrates 41 scientific tools, 35 of them served by 21 local microservices, and composes them into eleven domain workflows that cover the design of diverse RNA types. Capability-layer evidence gating (CapGate) evaluates every predictor on verified anchor sets, producing a capability map that drives two deterministic gates. A DecisionGate prevents unsupported actions, and a ClaimGate weakens or removes natural-language claims that exceed the evidence. Constraint navigation (GenNav) guides the generator through its control settings to produce candidates that satisfy multi-constraint requirements at a budget matched to the comparison.

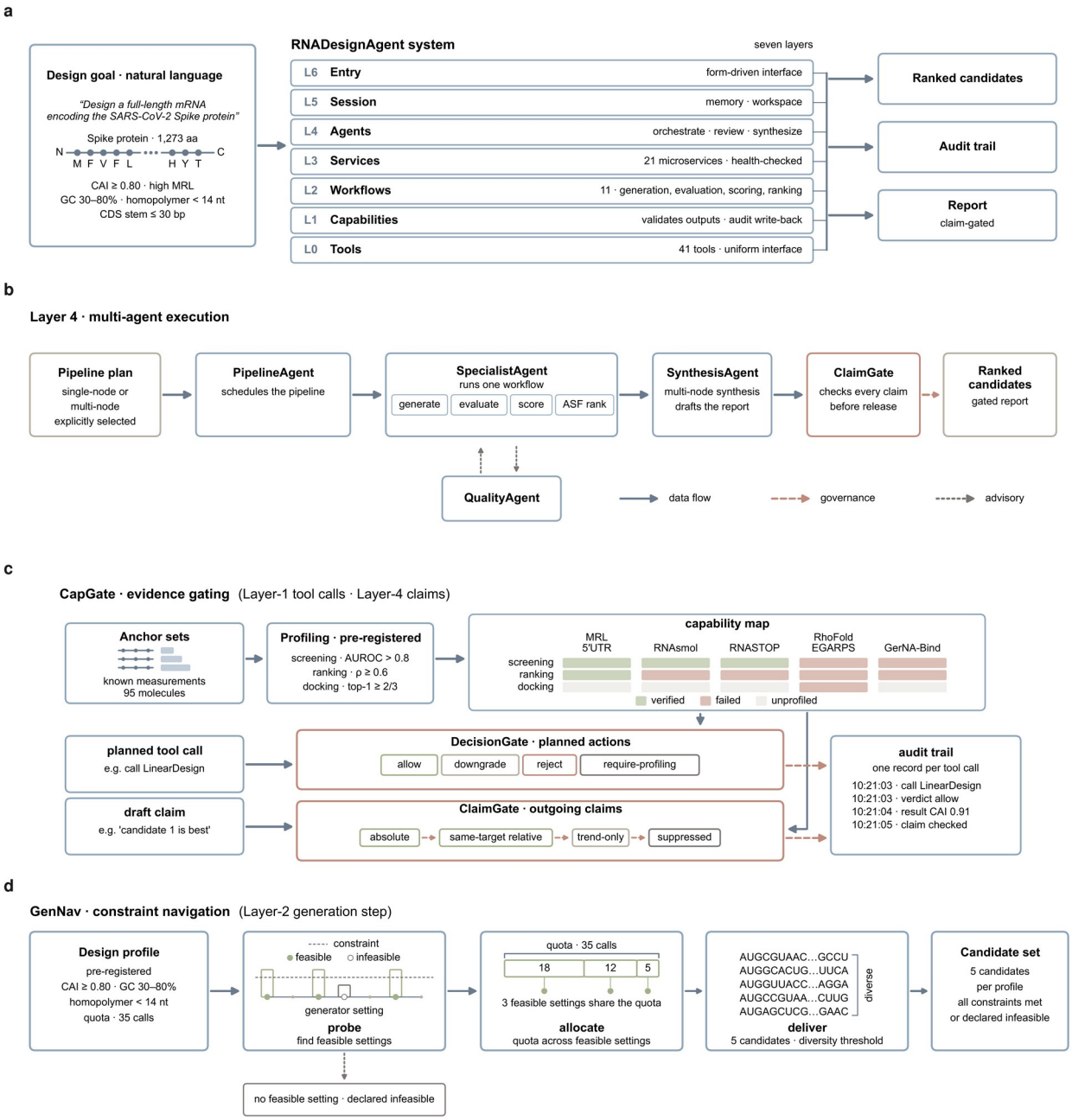


Fig. 1 | The RNADesignAgent architecture. (a) The RNADesignAgent system processes a natural-language design goal through its seven layers and returns ranked candidates, an audit trail and a claim-gated report to the user. (b) A detailed view of the multi-agent execution layer in panel a. The PipelineAgent schedules the pipeline, the SpecialistAgent runs each workflow and the SynthesisAgent drafts the report. The ClaimGate checks every claim before release, and the QualityAgent is advisory-only. (c) CapGate checks every tool call and every claim. Pre-registered profiling on verified anchor sets produces the capability map (verified, failed and unprofiled grades). The DecisionGate applies the map to planned actions (allow, downgrade, reject, require-profiling) and the ClaimGate to outgoing claims (absolute, same-target relative, trend-only, suppressed). (d) GenNav generates mRNA and aptamer candidates under a design profile. A feasibility probe first tests the generator's discrete settings. Generation then runs at the matched budget, every delivered batch meets the diversity threshold, and a target with no feasible setting is declared infeasible.

On RNADesign-210, a 210-task benchmark covering all eleven workflows, the system completed every task with biologically valid outputs. For example, designed coding sequences reached a median codon adaptation index (CAI) of 0.896, exceeding the wild-type median of 0.729 ($P = 1.1 \times 10^{-6}$)[18]. Capability profiling decertified all three deployed structure predictors on aptamer tasks but certified an expression predictor on measured ribosome-load anchors. On aptamer and full-length mRNA tasks, the resulting gates reduced false-positive claims to 0% and 7.7% from 54.5% and 33.3% ungated. On 25 mRNA targets, constraint navigation at

equal budget won the paired comparison against the static condition (20:6, McNemar P = 0.009). Transferred to the aptamer generator, the same mechanism won at low budgets (10:2, P = 0.039).

Results

Overview of the RNADesignAgent system

RNADesignAgent automates the design of diverse RNA types through a seven-layer architecture (Fig. 1a). The user states a design goal in natural language and receives designed candidate RNA molecules with their evaluation reports and a traceable record of the design process. The base layer (L0) exposes 41 scientific tools through a uniform, versioned interface, 35 of them served by 21 local microservices and the remaining six implemented within the system. The capability layer (L1) validates tool outputs and writes per-call audit records. The eleven domain workflows (L2) share a fixed sequence of generation, evaluation, scoring and ranking, and each addresses the design or evaluation of one RNA type. The service manager (L3) matches each workflow's declared requirements to microservices, health-checks them and starts whatever is missing. The multi-agent execution layer (L4) orchestrates, reviews and synthesizes workflow runs. Session memory and workspace management (L5) isolate each session's workspace and carry its result history across turns. The entry layer (L6), a form-driven interface, receives the user's natural-language design goal and returns the results (Fig. 1a). RNADesignAgent is built on the principle that the user must be able to reproduce and inspect the design process. Every design step is therefore executed by deterministic tools and recorded to the audit trail. The complete record of every task is returned to the user, who can trace which tools were called, with what parameters and under which gate verdicts, and replay the process.

The multi-agent execution layer (L4; Fig. 1b) comprises four cooperating agents. A PipelineAgent schedules multi-node pipelines, and per-workflow SpecialistAgents decide how to run the workflow and when to retry. A QualityAgent reviews outputs against per-workflow quality benchmarks, and a SynthesisAgent drafts multi-node reports whose claims pass through the ClaimGate before release.

Two mechanisms counter the three failure modes. CapGate (Fig. 1c) profiles each deployed predictor on verified anchor sets before use and enforces the resulting capability map at two deterministic gates. A DecisionGate guards the single tool-execution choke point through which every Layer-1 capability call passes, blocking unsupported predictor uses. A ClaimGate checks the SynthesisAgent's outgoing reports at Layer 4, downgrading or suppressing claims that exceed the evidence. GenNav (Fig. 1d) is a deterministic mode inside the Layer-2 workflows, implemented as a λ-curve navigator in mRNA coding sequence (CDS) design and a length–seed–sample navigator in RNA aptamer design[7]. It delivers candidate batches that satisfy multi-constraint profiles under a budget matched to the static condition.

RNADesignAgent generates biologically valid RNA molecules

We tested the biological validity of the system's outputs on RNADesign-210, the 210-task benchmark spanning all eleven workflows (Supplementary Tables S1 and S2). Every task completed successfully (210/210; Fig. 2a). On 20 wild-type protein targets, the delivered CDSs reached a median CAI of 0.896, above the wild-type sequences (0.729; Mann–Whitney $P = 1.1 \times 10^{-6}$, Cliff's $\delta = 0.91$). The wild-type sequences exceeded random synonymous sequences (0.675, n = 200; $P = 6.8 \times 10^{-7}$, $\delta = 0.67$), so the system reproduces and improves on the direction of natural selection on codon usage. The same delivered CDSs were substantially more structured than the two controls (folding energy per nucleotide −0.297 versus −0.051 and −0.047; $P \leq 6.8 \times 10^{-8}$, $\delta = -1.0$ for both comparisons). Designed sequences therefore lead in codon adaptation and stability simultaneously (Fig. 2b,c).

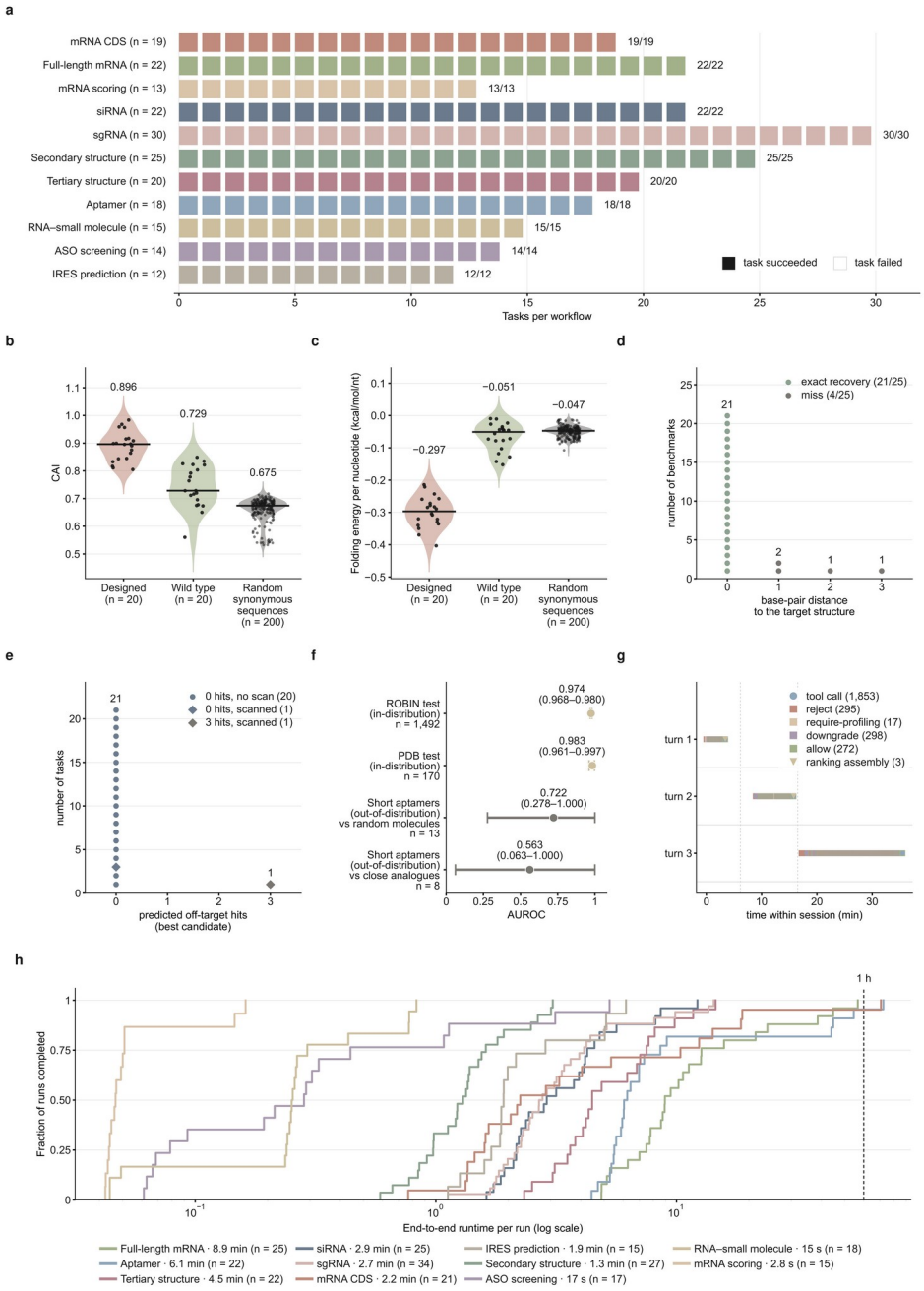


Fig. 2 | The system generates biologically valid outputs on RNADesign-210. (a) All 210 tasks succeed, one tile per benchmark task, grouped by workflow. (b) On the 20-target full-length mRNA set, designed CDSs reach a median CAI of 0.896 (n = 20), higher than wild type (0.729, n = 20) and random synonymous sequences (0.675, n = 200). (c) For the same three conditions, the designed CDSs have the lowest folding energy per nucleotide, and wild type and random sequences do not differ (P = 0.30). (d) Inverse folding recovers the exact target secondary structure in 21 of 25 benchmarks. (e) siRNA designs with 0 predicted off-targets for 21 of 22 tasks against the reference transcriptome. (f) The small-molecule predictor exactly reproduces its published evaluation to four decimals (95% CIs, 1,492 and 170 pairs, 0 failures). It does not generalize to out-of-distribution short aptamers. (g) A representative full-length mRNA design from the 241-run production campaign, with three user turns and 2,738 events, one row per user turn. (h) End-to-end runtimes by workflow (log scale, 1 h line) over all 210 benchmark tasks and 31 process-aware sessions. Medians range from 2.8 s (mRNA scoring) to 8.9 min (full-length mRNA design), 70.1% complete within 5 min, 89.2% within 10 min and 239/241 within 1 h, the exceptions being CDS design at 70.7 min and aptamer design at 72.3 min.

Inverse folding reproduced the exact target secondary structure in 21 of 25 benchmarks (Fig. 2d). The four misses were near-exact, each differing from its target by one to three base pairs, and no candidate in the ranked set exactly reproduced any missed target, so the shortfall lies in the generation tool rather than the system's ranking[8]. On 22 siRNA design tasks, 21 delivered a best-ranked candidate with zero predicted off-targets (Fig. 2e). The only exception is due to the benchmark input rather than the scanner. Its synthetic target is an exact fragment of the human long non-coding RNA locus DDX11L2, so the three predicted hits are the guide's own

locus and its two identical paralogous copies in DDX11L16, true genomic occurrences rather than false partial matches.

The small-molecule binding predictor matched its capability profile exactly (Fig. 2f). Through the deployed service it reproduces its published evaluations exactly, with an area under the receiver operating characteristic curve (AUROC) of 0.974 and 0.983 on the ROBIN and Protein Data Bank (PDB) test sets (95% confidence interval (CI) 0.968–0.980 and 0.961–0.997; n = 1,492 and 170 pairs), but shows no reliable power on short aptamers outside the training distribution (0.722 and 0.563 against random molecules and close analogues, both intervals spanning chance)[19,20]. The remaining six workflows are reported in Supplementary Notes 1 and 2 (Supplementary Figs. S1–S3).

Every run produces a complete audit trail, with one record per tool call. The 241 production runs, covering all 210 benchmark tasks plus 31 regression sessions entered through the user form, left 24,897 audit events. Of these, 19,848 were tool calls; 4,862 were gate verdicts; and 187 were assemblies of ranking objectives. Fig. 2g traces every event class call-by-call in a representative full-length mRNA session from the same trail (1,853 tool calls, 882 gate verdicts and 3 ranking assemblies). A full design run completes in minutes to roughly an hour. Across the same 241 runs, the pooled median runtime was 2.5 min, with 239 of 241 finishing within 1 h (Fig. 2h).

Current scientific agents exhibit three systematic failure modes

Current scientific agents optimize predictor scores without validating the predictor's power, report their own results without verification, and evaluate improvements under unequal sampling budgets. These practices produce three systematic failure modes, which we quantify on the system itself (Fig. 3). Uncalibrated trust corrupts the agent's actions, claim overreach corrupts its reports, and confounded improvement corrupts its own evaluation.

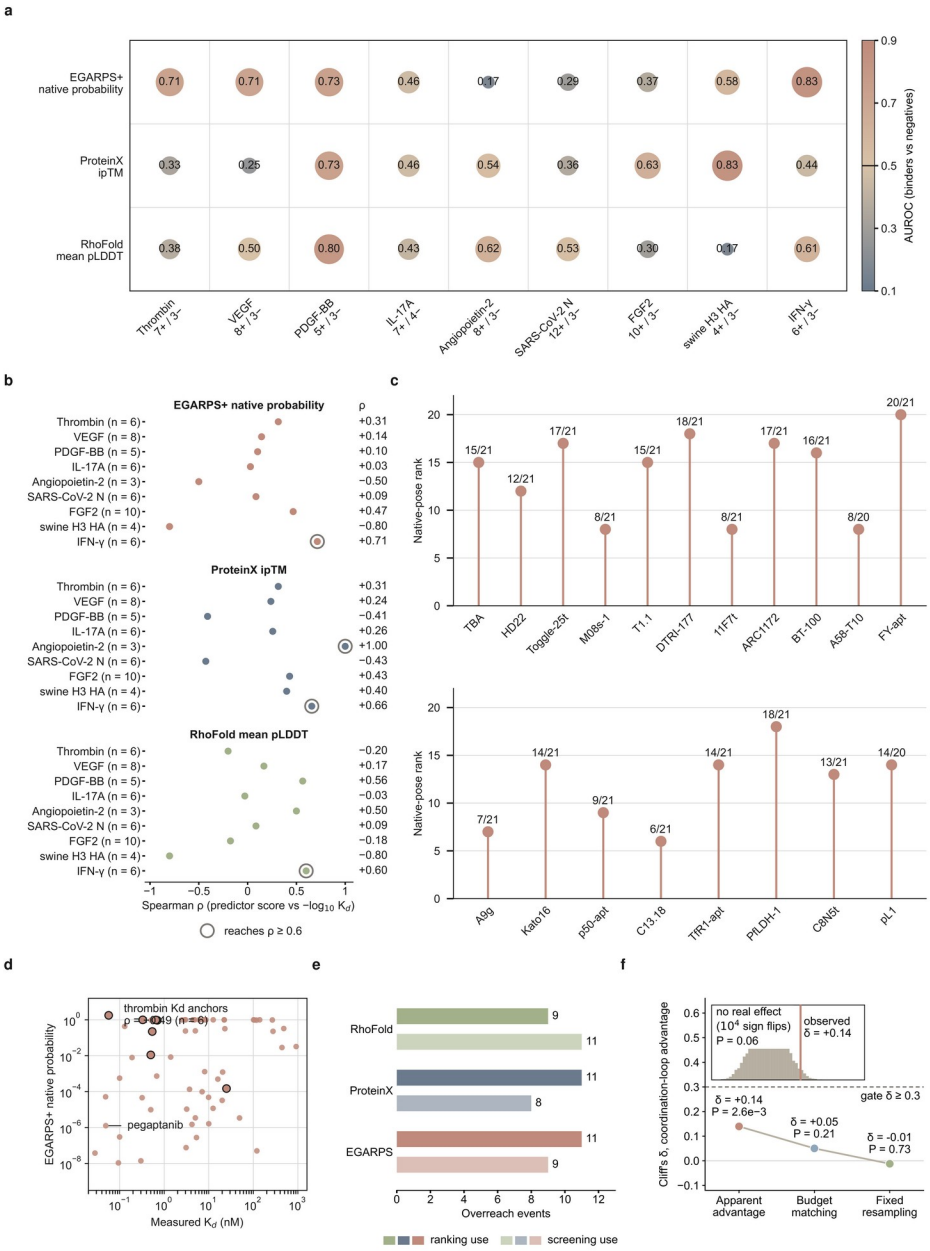


Fig. 3 | Current scientific agents exhibit three systematic failure modes. (a) Screening power of three structure predictors across nine RNA–protein targets. On the 95-sequence anchor set, only 2 of 27 evaluations pass the AUROC > 0.8 gate, ProteinX on swine H3 HA (0.83, CI 0.33–1.0) and EGARPS on IFN-γ (0.83, CI 0.50–1.0), but all 27 95% CIs include 0.5. (b) Ranking power of the three structure predictors across the same nine targets. Only 4 of 27 evaluations reach $\rho \geq 0.6$, all three predictors on IFN-γ and ProteinX on Angiopoietin-2. (c) Docking power of EGARPS on 19 solved RNA–protein complexes across 14 protein targets. The native pose never ranks top-1 (0/19; gate at ≥ 2/3 of complexes) and falls below the median decoy in most complexes. (d) Calibration of predicted scores to measured Kd. The scores do not separate binders from negatives (AUROC 0.52, 95% CI 0.39–0.65, 62 binders and 28 negatives), the medians of binders, literature negatives and random negatives interleave (0.22, 1.4×10^{-3} and 2.2×10^{-2}), and the scrambled TBA negative (score 0.98) outscores the binder pegaptanib (Kd ~1.3×10^{-6}). (e) Claim overreach in the ungated agent, plan-level events per predictor in ranking and screening use. (f) Confounded improvement. The coordination loop's apparent advantage over the single-run baseline (Cliff's $\delta = +0.14$, paired Wilcoxon $P = 2.6 \times 10^{-3}$) falls under budget matching ($\delta = +0.05$, $P = 0.21$) and disappears under fixed resampling at the same mechanics and budget ($\delta = -0.01$, $P = 0.73$).

The RNA aptamer design workflow ranks candidates by the scores of three structure-based predictors, EGARPS+ native probability, ProteinX interface confidence (ipTM) and RhoFold mean per-residue confidence (pLDDT)[21–24]. To quantify uncalibrated tool trust, we profiled all three on verified anchor sets. The affinity set holds 95 sequence entries spanning nine protein targets (67 binders, 62 of them with a measured dissociation constant (Kd), and 28 negatives). The docking set holds 19 solved RNA–protein complexes across 14 independent protein targets. Each predictor–target combination was graded by thresholds for screening (AUROC above 0.8), ranking (Spearman ρ at least 0.6) and docking (native pose top-1).

None of the three showed reliable power in any capability. For screening, only 2 of the 27 predictor–target combinations pass the gate, ProteinX on swine H3 HA and EGARPS on IFN-γ, but all 27 confidence intervals span chance (Fig. 3a). FGF2, with the widest Kd range, stays sub-gate for all three predictors. For ranking, 4 of 27 combinations reach $\rho \geq 0.6$, all three predictors on the IFN-γ anchors ($\rho = 0.71$, 0.66 and 0.60, $n = 6$) and ProteinX on Angiopoietin-2 ($\rho = 1.0$, $n = 3$) (Fig. 3b). The three IFN-γ intervals span zero, and those anchors are one parent aptamer and its in-silico mutants spanning under one order of magnitude of Kd, while the Angiopoietin-2 estimate rests on three points whose scores flip between ~0 and ~1 across stochastic complex realizations. For docking, the native pose ranked top-1 in 0 of 19 complexes, falling at ranks 6–20 of 21 candidates (Fig. 3c). The Kd analysis also gave the same conclusion (Fig. 3d). The thrombin Kd anchors ($n = 6$) reached $\rho = 0.49$, below the ranking gate, and the predictor did not separate binders from negatives better than chance. The three predictors are therefore not reliable optimization objectives, and an agent that optimizes these scores may raise them without obtaining a better RNA molecule.

The second failure mode, claim overreach, occurs because agents report their own results without verification (Fig. 3e). We ran an ungated agent on a 45-task audit set of design-reporting tasks. It made false-positive claims at a rate of 54.5% (30 of 55 claims, 95% CI 41.5–67.0%), and its plans or statements exceeded what the evidence supported 59 times, at 8–11 events per predictor–use combination (ranking or screening). The failure occurs only where the evidence is untrusted. The 24 deterministic tasks produced 23 claims, every one verified true by exact recomputation. The 21 untrusted-predictor tasks were scored by the three predictors profiled above and produced 32 claims, 30 of them false. The agent declares the predictor's top scorer the best binder, and the held-out Kd ground truth shows otherwise. The failure is not caused by the task construction either, because the original and rule-extended tasks fail at the same false-positive rate (55.6% versus 53.6%).

The third failure mode, confounded improvement, emerged in the system's earlier results (Fig. 3f). RNADesignAgent's earlier coordination loop was run across the full aptamer benchmark. We compared it against a single-run baseline, the same run given the loop's budget, and a fixed-resampling control that replaces its between-round analysis with content-free regeneration. On the uncalibrated EGARPS-component metric, the loop showed an apparent advantage (Cliff's $\delta = +0.14$), because each coordination round doubles the evaluated pool. The advantage falls to $\delta = +0.05$ at the loop's budget and disappears ($\delta = -0.01$) in the fixed-resampling control. Confounded improvement is not specific to this loop, but a general risk whenever improvements are evaluated under unequal sampling budgets. We therefore enforce budget matching on every comparison in this paper.

Capability gating removes unsupported claims while preserving true ones

CapGate prevents uncalibrated trust and claim overreach through two deterministic gates that the capability map drives (Fig. 1c). The DecisionGate checks planned actions and forbids those the capability map does not support, such as ranking candidates by a predictor with no measured ranking power. The ClaimGate checks natural-language claims before release and downgrades those that exceed the evidence through a fixed sequence of scopes (absolute, same-target relative, trend-only, suppressed). Every call to a governed predictor is checked against the map at the system's single tool-execution choke point, so an unevidenced call is rejected before it executes rather than flagged afterwards. A predictor without profiling evidence stays blocked by default until evidence exists.

The evidence that capability profiling produces determines how reliable the gates can be. Profiling measures each predictor on ground-truth anchor sets, and the resulting grades form the capability map. Two properties of this evidence are important, two-sidedness and fidelity to the deployed tool. Profiling withholds unsupported power and certifies supported power, so the capability map records what the predictors can do, not only what they cannot. Both sides appeared when we profiled the system's own predictors. The GEMORNA 5′ untranslated region (5′UTR) mean ribosome load (MRL) predictor passed on all three libraries of a published massively parallel translation assay (706 anchors; Fig. 4a), but the 3′ untranslated region (3′UTR) stability predictor is unprofiled and the DecisionGate blocks it from ranking use[25,26]. These grades become the gates' per-request verdicts, re-measured as a decision matrix of twelve combinations and verified live at the tool layer (Fig. 4b). Ranking by the MRL predictor is allowed on all three certified libraries, downgraded rather than

blocked on an unprofiled target family, and blocked for the 3′UTR-stability predictor, so on certified predictors the gates are selective rather than prohibitive.

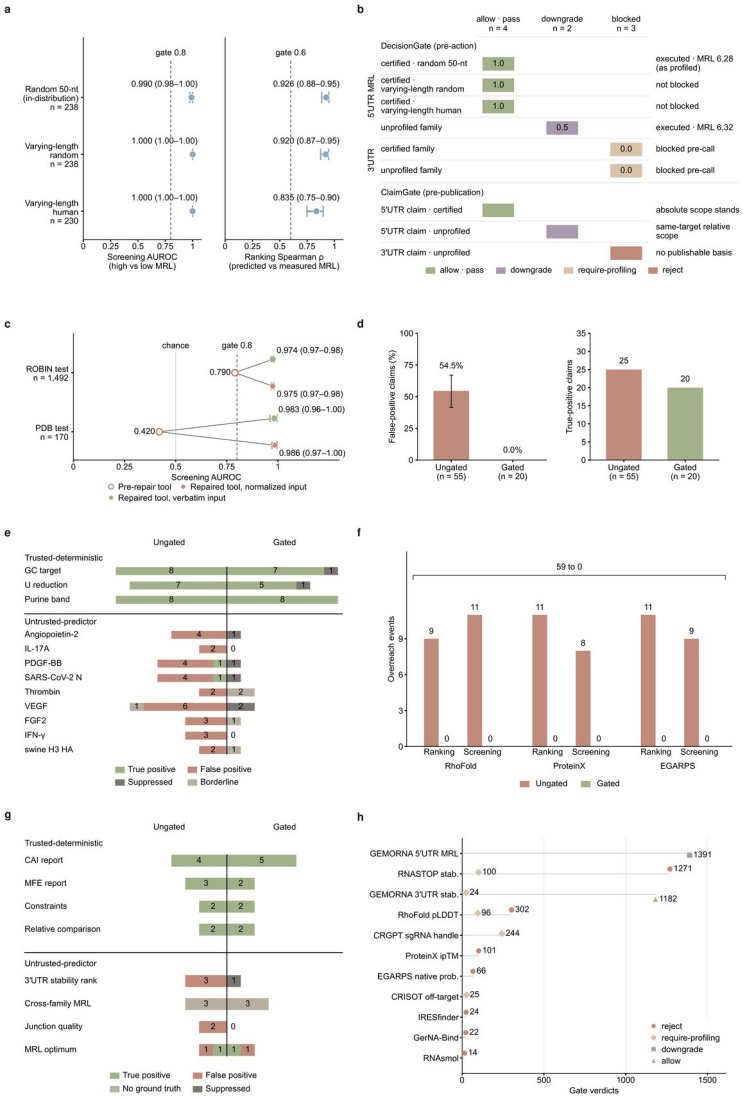


Fig. 4 | Capability gating removes unsupported claims while preserving true ones. (a) Capability profiling of the expression predictor (GEMORNA MRL) on three 5′UTR libraries, the random 50-nucleotide library and two never-trained varying-length libraries. Screening AUROC (gate 0.8) and ranking ρ (gate 0.6) are each computed on a 120-anchor subset of each library (screening 60 positive and 60 negative). The predictor passes on all three (screening AUROC 0.990–1.000, ranking ρ 0.835–0.926, CIs 0.75–0.95), with every anchor of the three libraries scored (706/706). (b) Gating on the certified MRL predictor. Ranking requests are allowed on the three certified libraries (confidence 1.0), downgraded on an unprofiled target family (confidence 0.5) and blocked for the unprofiled 3′UTR stability predictor. Absolute claims pass on a certified library, downgrade to same-target relative scope on the unprofiled family and are rejected for the unprofiled one. The right columns give the recorded confidence and the measured tool-layer outcome. (c) Measuring the deployed tool. The pre-repair tool scored 0.790/0.420 (matching the registered incident record to ≤ 0.0094). After the repair, the tool was re-verified live on all 1,662 pairs and reproduces its published performance. (d) The ungated and gated conditions on the 45-task audit set, identical except for the gates. (e) Claim verdicts by task group. The ungated condition has 25 true and 30 false positives, the gated 20 and 0, with 7 suppressed and 4 borderline. A printed 0 marks a group and condition with no adjudicated claim. (f) Plan-level overreach events per predictor in ranking and screening use. (g) The same per-group view on the mRNA audit set. The ungated false positives come from the 3′UTR-stability (3), junction (2) and MRL-optimum (1) task groups. (h) Gate verdicts by predictor across the 241-run production campaign. Of the 4,862 verdicts, 1,800 reject, 489 require profiling, 1,391 downgrade and 1,182 allow.

Profiling must also measure the deployed tool rather than trust the publication (Fig. 4c). The small-molecule predictor RNAsmol initially failed to reproduce its published performance even on its own test sets, a degradation that only profiling caught[19]. Comparing the port's preprocessing against the upstream original revealed two differences, a case collapse in the model's case-sensitive RNA encoding and the correction of an upstream bug in the adjacency construction. The difference that looked obviously fatal was harmless. The one that looked like an improvement destroyed the predictor, because the checkpoints had been trained with the bug

and correcting it changed their input distribution. After the repair, the deployed tool reproduced its published performance (Fig. 4c). Tools can silently diverge from their papers, and only measurement reveals where the damage lies.

We then tested the gates in paired two-condition experiments on aptamer and mRNA tasks, identical except for the gates (Supplementary Methods M1). The aptamer tasks are the 45-task audit set of the claim-overreach analysis. On this set, gating reduced the false-positive claim rate from 54.5% (30 of 55) to exactly zero (0 of 20; Fig. 4d,e). Representative ungated claims and their gated versions are tabulated in Supplementary Table S3. Plan-level overreach events fall from 59 to zero (Fig. 4f). The decision gate blocked all 71 plan requests, so the gated agent never even obtained the scores it could not use. The guarantee of zero false claims had a measurable cost. True-positive retention was 80.0% (25 true positives, 20 kept), below the 90% gate. Two true claims fell to the over-conservative uniform basis rule, two wrong-length sequences were abstained before their false claims could be published, and two lucky true positives on untrusted predictors could not be kept, with one true positive rescued.

The mRNA tasks are a 20-task set built from real full-length design-reporting tasks (Fig. 4g). Gating reduced the false-positive rate from 33.3% (6 of 18) to 7.7% (1 of 13), a 76.9% relative reduction, and kept all 12 true positives. Plan-level overreach falls from 5 to 0. The retention contrast with the aptamer side is a governance effect. The full-length mRNA workflow no longer ranks on the unprofiled 3′UTR predictor, so the expression claims on unprofiled target families that previously reduced the true-positive count are no longer made. The decision gate's five blocks on the stability and junction ranking plans left the model no evidentiary path, and it published nothing rather than fabricate a claim. The one remaining false positive came from the MRL predictor ranking the wrong candidate first on the weakest certified library.

The gates show the same behaviour at production scale (Fig. 4h). In the 241 production runs, the decision gate issued 4,862 verdicts across eleven predictors. The rejects concentrate on the predictors that failed profiling and the require-profiling verdicts on the unprofiled ones, while the downgrades all fall on the certified MRL predictor, whose grade transfers to unprofiled target families with reduced confidence. Every allowed request is a call to the 3′UTR-stability predictor, only to report its score, never to rank candidates. No ranking request is allowed, because no verified ranking grade exists for these tasks.

Constraint navigation efficiently generates RNA molecules that meet the design requirements

The agent's task is multi-constraint batch delivery. The generator's settings already determine any single quality metric such as codon adaptation or folding energy, so single-metric design leaves nothing for an agent to decide. For each target, the system must deliver a ranked batch of candidates that jointly satisfy a constraint matrix. GenNav achieves this goal at the generation step (Fig. 1d). It first tests a few settings to find which can meet the constraints, and only then commits its quota of generation calls, identical in every comparison. Any two candidates in a delivered batch differ in a minimum fraction of positions, a diversity threshold that prevents near-identical designs. A target with no feasible setting is reported as infeasible with its failure reason rather than silently omitted.

A first attempt placed navigation in the selection step, generating a single fixed pool of 315 full-length candidates per target (35 CDSs × 9 UTR combinations) and then filtering and ranking it. It fails for reasons inherent in generation (Fig. 5a). Across the five targets, the number of candidates passing all constraints was 306 of 315 for EPO, 297 for IL-2, 144 for insulin, 27 for A1AT and zero for GFP under the default codon table. No selection-step ranking can recover candidates the generator never produced. Even with 144 passing candidates, insulin's delivered candidates all come from one generator setting, so they fall below the diversity threshold. Navigation therefore moved to the generation step.

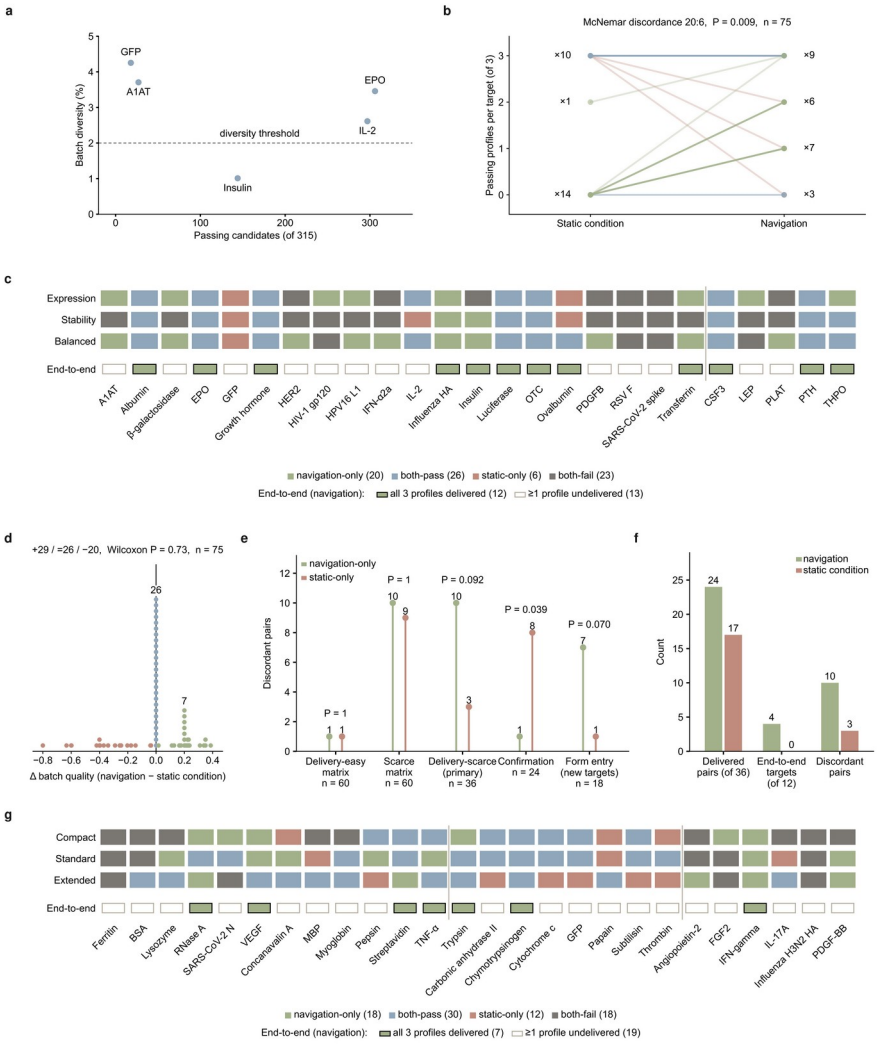


Fig. 5 | Constraint navigation on full-length mRNA and aptamer design. (a) The selection-step attempt fails for reasons inherent in generation. Insulin's batch diversity is 1.01% (all candidates from one λ setting), and the GFP pool passes 18 of 315 candidates under the species-matched codon table. (b) The mRNA primary comparison over 25 targets at 35 matched candidates, per-target passing-profile counts under the two conditions connected by lines coloured by the discordance direction (sage navigation-favouring, terracotta static-favouring, grey blue concordant), n = 75 pairs. (c) Delivery outcomes for the 25 mRNA targets and 3 profiles, with 20 navigation-only, 26 both-pass, 6 static-only and 23 both-fail. The end-to-end row marks the 12 targets delivered in all three profiles. (d) Batch quality of the delivered pairs. The rate of targets delivered in all three profiles (0.48) misses the legacy 0.8 gate (static pool 1.00), and at the met constraint-rate criterion (0.96 ≥ 0.9) batch quality does not differ (Δ +29 / =26 / −20, P = 0.73), with 0 budget violations and mean navigation spend 23.1 of 35 candidates. (e) The aptamer campaigns at 48 matched candidates, exact McNemar per campaign. Both conditions deliver under the delivery-easy matrix (1:1, 56/60 both-pass, navigator spend 32.7 of 48), the scarce matrix is balanced (10:9), the navigator delivers more pairs on the 12 delivery-scarce targets (10:3, P = 0.092), the static condition more on the confirmation set (1:8, P = 0.039), the navigator more on the form-entry targets (7:1, P = 0.070). Pooled over 26 targets, navigation delivers 48 pairs to 42 (18:12, P = 0.36). (f) The 12 delivery-scarce targets. Navigation delivers 24 of 36 pairs and the static condition 17, and only navigation delivers all three profiles (4 targets). (g) Delivery outcomes for the 26 aptamer targets and 3 profiles, with 18 navigation-only, 30 both-pass, 12 static-only and 18 both-fail pairs (including the Ferritin-class zero-candidate targets). The end-to-end row marks the 7 targets delivered in all three profiles.

Across 25 targets, navigation wins the paired comparison against the static condition. Each target is designed under three profiles emphasizing expression, stability and balance, giving one paired outcome per target and profile. In 26 of the 75 pairs only one condition delivers a passing batch, navigation in 20 and the static condition in 6 (exact McNemar P = 0.009; Fig. 5b,c). Navigation spends less than its quota and delivers passing batches in 46 of the 75 pairs, the static condition in 32. Twenty-two of the 25 targets receive at least one passing batch from navigation. The quality of the delivered batches does not differ between the two conditions, so navigation's advantage is delivering more pairs (Fig. 5d).

We also tested the navigation mechanism on RNA aptamer design. CapGate blocks every structure predictor from ranking candidates, because profiling found no reliable ranking power in any of them. The diffusion generator's per-setting quality is mostly sample-level noise, and each probe sample adds only about 15 percentage points of signal[27]. A campaign on the 20 aptamer targets shows where the noise matters (Fig. 5e). Under a delivery-easy constraint matrix both conditions deliver successfully, and the navigator stops early without delivering less. On the 12 most delivery-scarce targets, only one condition delivers a passing batch in 13 of the 36 pairs, the navigator in 10 and the static condition in 3. The navigator delivers in all three profiles on 4 targets and the static condition on none (Fig. 5f,g). Notably, this advantage is not statistically significant ($P = 0.092$), because the comparison is based on only 13 pairs. Across all 26 aptamer targets, including six that entered through the production form entry, navigation delivers in 48 pairs to the static condition's 42 (Fig. 5g). To determine when navigation is suitable, we further conducted a 39-target budget sweep (Supplementary Fig. S4). Navigation is decisively better when the quota is scarce relative to the target's difficulty (10:2, $P = 0.039$). Both conditions deliver fully at moderate quotas, and navigation spends about a third fewer generator calls to do so. The static condition wins when the quota is loose, because its exhaustive per-setting depth delivers more pairs (12:30, $P = 0.008$).

The system outperforms bare-model and tool-using agent baselines

To evaluate the performance of RNADesignAgent, we compared it against two baselines, a bare language model and a generic tool-using agent, on the 31-session process-aware benchmark. A bare-model condition used the same underlying model at temperature 0, prompted zero-shot and few-shot, with no tool access. A generic ReAct-style tool-using loop was given the same 41 tools through the same audited executor[28]. The loop was tested at the system-matched budget and at disclosed 3× and 10× budgets. A turn counts as delivered only if its output passes the workflow's validity checks and, on the two graded structure tasks, matches the system's best candidate on that turn. A baseline's fold-back base-pair distance to the target secondary structure, or its alignment template-modelling (TM) score to the target tertiary structure, must match or beat what the system's best achieved on that turn. Natural-language claims were scored separately and blindly.

The system delivered 96 of 96 turns (Fig. 6a). The bare model delivered 71 zero-shot and 69 few-shot, and the tool-using loop delivered 7 at the system-matched budget, 44 at 3× and 71 at 10×. No turn was delivered by a baseline and missed by the system, and every comparison is significant (exact McNemar, $P < 10^{-6}$, $n = 96$). The baselines fail for different reasons (Fig. 6a,b). The bare model fails on capability, with zero-shot answers below the system's best on 14 turns and invalid on 11 turns. The loop's failures at the matched and 3× budgets are almost entirely budget exhaustion without an answer (86 and 42 turns). Every step of the loop is a model call, so at the matched budget the two calls per turn the system spends on review cover a single tool round trip, leaving nothing for a second attempt. At the 10× budget the loop's remaining failures are mostly quality failures (16 turns below the system's best and 9 still exhausting budget). The two baselines deliver the same number of turns while winning different workflows. The loop outperforms the bare model where computation is essential, on tertiary-structure inverse folding, sgRNA design and coding-sequence and full-length design. The bare model outperforms the loop where direct reasoning suffices. On antisense windows, the loop exhausts its budget on accessibility scans of a long transcript and never submits, and on secondary-structure targets of 15–23 nucleotides its submissions miss the system's exact recovery by one to six base pairs.

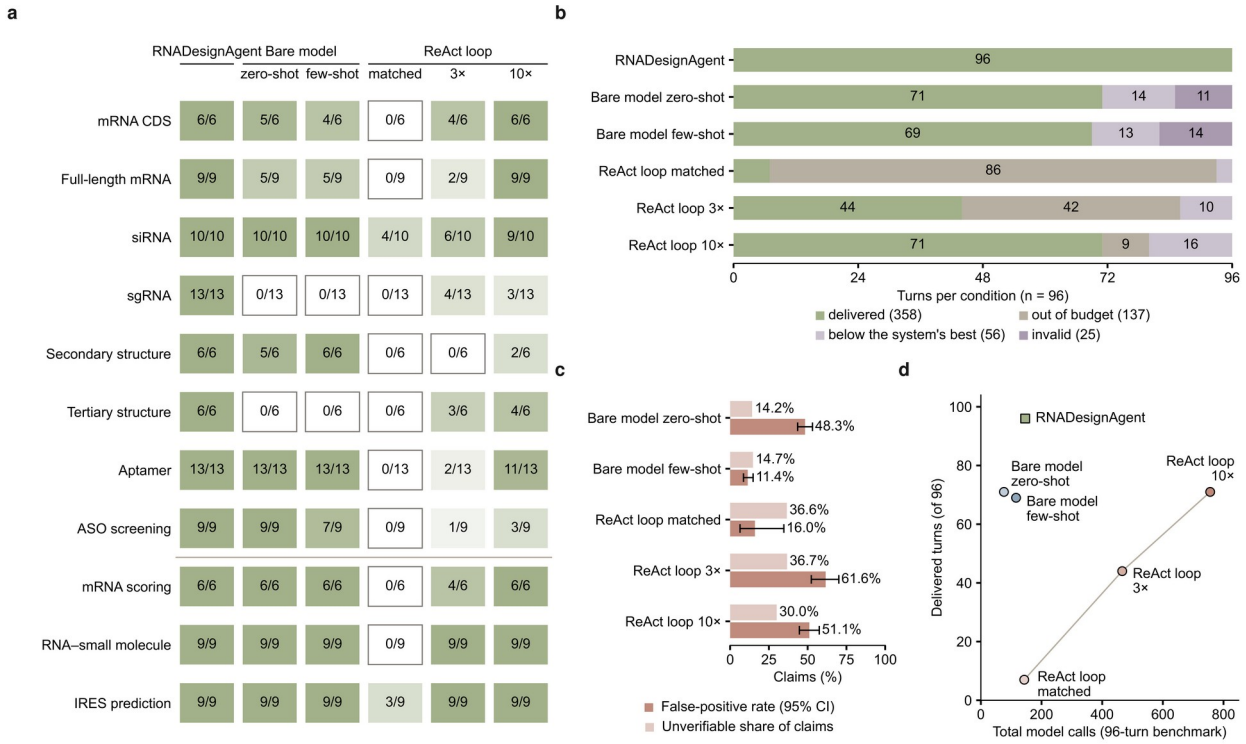


Fig. 6 | The system outperforms bare-model and tool-using agent baselines. (a) Delivered turns for the eleven workflows under six conditions, with the system, the bare model zero-shot and few-shot, and the same-tool loop at matched, 3× and 10× model-call budgets. Design workflows are above the divider, evaluation workflows below. (b) Failure modes of the 96 turns per condition, delivered, out of budget, below the system's best and invalid. (c) Baseline claims, false-positive rates (Wilson 95% CIs) and unverifiable shares. (d) Total model calls and delivered turns for each condition. The system is at the upper left, and the loop reaches the bare model's delivery rate only at the 10× budget. The two bare-model markers share 96 calls and are drawn at a symmetric ±20-call offset for visibility, all other markers at exact coordinates.

The system and the two baselines differ even more in their claims than in their delivery (Fig. 6c). The system makes no claims from single workflows, because its claim gate operates only on the synthesis reports of multi-node pipelines, while both baselines make claims freely. The bare model's recomputable claims were false positive at 48.3% zero-shot but only 11.4% few-shot, because the worked examples guide the arithmetic it can do without tools. The tool-using loop's claims were false positive at 51.9%, and 30–37% were unverifiable. The ungated-control rate of 54.5% therefore generalizes from our own ungated pipeline to a generic agent.

The system also used far fewer model calls (Fig. 6d). The system completed the entire benchmark on 146 model calls, all of them quality review. To reach the bare model's delivery rate, the loop needed 756 model and 659 tool calls at the 10× budget, 5.2 times the system's model spend. On the deterministic workflows (sequence scoring, small-molecule prediction and antisense screening) the system uses no model calls at all, so the system-matched budget for those eight sessions is exactly zero, and a loop whose every step is a model call cannot even take its first action. All 24 turns there end in recorded budget exhaustion with no answer. The 3× and 10× budgets guarantee a minimum of two calls per turn, so the loop can finally take its first action on these workflows. At 3× it delivers 9 of 9 small-molecule turns, 4 of 6 sequence-scoring and 1 of 9 antisense. Tool access by itself therefore delivers valid outputs only where computation is essential and adds cost on the other workflows. The system's advantage, every turn delivered at a fifth of the loop's model cost with no false claims, comes from its deterministic workflows and governance, not only from the model.

Discussion

In this work, we developed RNADesignAgent, a multi-agent system for RNA design that addresses the three systematic failure modes of current scientific agents. Uncalibrated tool trust, claim overreach and confounded improvement are not theoretical risks but patterns we measured. CapGate prevents unsupported actions and claims, profiling each predictor on verified anchor sets before use and checking every outgoing report against

the evidence. GenNav prevents confounded improvement, generating candidates at a budget matched to the comparison, and we enforce budget matching on every comparison in this paper. Profiling found no reliable power for any predictor we tested on the aptamer side, in screening, ranking or docking. It certified the expression predictor on measured ribosome-load anchors. CapGate removed almost all unsupported claims on aptamer and full-length mRNA tasks while preserving most true ones. On mRNA tasks, GenNav outperformed the static condition at equal budget. On aptamer tasks, GenNav is decisively better when the quota is scarce, spends fewer calls at moderate quotas and gives way to the static condition when the quota is loose.

The reliability of an agent comes not only from its language model but from the anchor sets, capability map, gates and audit trail around it. Both actions and claims are gated because language can exceed the evidence. Both outcomes and budgets are controlled because more sampling can mimic improvement. On multi-step pipeline runs, the ClaimGate rejected five of the twelve claims drafted by the synthesis agent as unsupported. The system was also compared against agent baselines under matched model-call budgets. Even a tool-using loop with the same underlying model and the system's tools delivered almost nothing at the matched budget. The loop's claims were ungated and failed at the rate of the ungated control.

RNADesignAgent currently has two main limitations. GenNav is implemented end-to-end in only two of the eleven workflows, full-length mRNA design and RNA aptamer design, limited by the availability of suitable generation tools. As design tools for the remaining RNA types become available, GenNav can be extended to the corresponding workflows. In addition, the anchor sets that calibrate the predictors in these two workflows are verified but limited in scale. Richer anchor sets will improve the calibration of the predictors and thus the reliability of their predictions.

RNADesignAgent automates the design of diverse RNA types, and its results can be verified rather than merely reported, with CapGate and GenNav as two of its mechanisms. The discipline behind them, profiling before optimizing, gating before claiming and matching budgets before comparing, applies to other scientific agents. For RNA design, the system turns a fast-growing toolbox into a single automated process driven by natural language, making design more efficient and easier for biologists.

Methods

System architecture of RNADesignAgent

RNADesignAgent consists of seven layers. L0 (tools) contains 41 scientific tools, each with a machine-readable definition of its inputs and outputs. Thirty-five of them are served by 21 local FastAPI microservices, each of which runs one external software package or the system's own sequence metrics in its own runtime environment with local model weights. The remaining six are deterministic implementations within the system. L1 (capabilities) validates the output of every tool call against the tool's definition and records each call in the run's audit trail. L2 (workflows) contains the eleven workflows, with the design workflows all following the same process of generation, evaluation, scoring and ranking (Supplementary Tables S1 and S2). L3 (service management) matches each workflow's declared capability requirements to microservices and starts any missing service automatically. L4 (multi-agent execution) runs the workflows through four cooperating agents. A PipelineAgent schedules multi-node pipelines, a SpecialistAgent executes each workflow, a QualityAgent reviews the workflow's outputs and a SynthesisAgent drafts multi-node reports. L5 (session and workspace) assigns each session its own workspace directory, where input and output files are kept, and preserves earlier results so that later turns can build on them. L6 (entry) provides a form-driven command-line interface (CLI) and a FastAPI interface. The user selects the workflow explicitly, and every user-supplied parameter is validated against the workflow's input definition.

The multi-agent layer

The multi-agent layer comprises four agents, each with a single specific role. A PipelineAgent schedules the nodes of multi-node pipelines, handling input binding, critical failures and final synthesis, and makes no LLM calls. A SpecialistAgent executes each workflow, obtains a quality review and decides whether to retry. Retries are limited to hard failures, a run that returns no candidates or one whose tool calls fail. Both conditions are

objective and reproducible, unlike a reviewer's dissatisfaction. A QualityAgent reviews each workflow's outputs against per-workflow quality benchmarks and produces an advisory report, except in the three deterministic evaluation-only workflows. The report is shown to the user and cannot trigger a retry. A SynthesisAgent drafts multi-node reports, and every claim passes through the ClaimGate before release. The advisory review and every agent's contribution are measured in a three-condition advisory experiment and a registered per-agent ablation (Supplementary Note 3; Supplementary Figs. S5 and S6). The LLM used by the system is DeepSeek (deepseek-v4-pro via the official API)[29].

Candidate ranking by the achievement scalarizing function

Every workflow with two or more optimization objectives of different kinds ranks its candidate sequences through a unified implementation of the achievement scalarizing function (ASF)[30]. Objective values are min–max normalized to [0, 1] across the candidate sequences, and candidates are ordered by an augmented Chebyshev ASF against a reference point r, defined as $\mathrm{ASF}(s) = \max_i \{w_i (r_i - s_i)\} + \rho \sum_i |w_i (r_i - s_i)|$ with $\rho = 10^{-3}$, which ensures that the minimizer is Pareto-optimal. When a workflow's weight parameters cover all objectives, they are normalized and used. Otherwise the objectives are weighted equally. The user can supply the reference point using original values. The default is the ideal point, the best value on each objective among the candidates. The ordering is always accompanied by a Pareto frontier on the original objective values. Missing data are handled by three policies. A dimension missing for every candidate, for example when CapGate blocks a predictor, is dropped from the ranking. A dimension missing for one candidate is assigned the worst value measured on that dimension, so a failed evaluation never helps that candidate. A user reference point that covers only some dimensions restricts the ranking to those dimensions and is flagged in the summary. Six of the eleven workflows rank through this implementation, with their objectives listed in Supplementary Table S1. The remaining five are single-objective, single-candidate or single-score tasks and rank directly without the ASF. An ASF value is meaningful only within its candidate pool, because ranks can change when the pool changes and the ideal reference can be attained (ASF = 0). Cross-pool comparisons therefore never compare raw ASF values and instead use shared fixed reference points, per-objective components or the volume dominated by a pool's Pareto frontier.

Capability profiling and evidence gating (CapGate)

CapGate consults the capability map, a table included with the system and built from the verdicts of the capability audits. For each predictor, task type and anchor set, the map records one of four capability grades (screening, ranking, docking or none), the underlying evidence and a confidence value. The CapabilityProfiler computes each grade by evaluating the predictor on anchor sets and applying pre-registered thresholds. The DecisionGate checks planned actions and forbids those the capability map does not support. The ClaimGate checks natural-language claims before release and downgrades those that exceed the evidence through a fixed sequence of scopes (absolute, same-target relative, trend-only, suppressed). CapGate reviews the governed predictors against the capability map at the single tool-execution choke point. A call whose grade is failed or unprofiled is blocked and its dimension is dropped from the ranking, so the run completes instead of stopping with an error. When a predictor is used on a target outside its profiled anchor sets, each capability is assigned the worst grade measured on any of those sets. Each anchor in a set is labelled binder, negative or unrankable. Before use, the system checks the set's internal consistency, its negatives and coverage of at least 4 binders and 3 negatives per target. mRNA anchors follow the same pattern, with measured MRL values in place of affinities. For the screening grade, the values are split into high and low at pre-registered pool extremes. For the ranking grade, a subset stratified by value is drawn from each library.

Constraint navigation and budget-matched delivery (GenNav)

GenNav comprises four components, a feasibility probe, delivery for each design profile, a diversity threshold and a matched generator configuration. A probe first tests the generator's discrete settings on the target and identifies the feasible ones before any generation budget is spent. If no setting is feasible, the target is reported as infeasible. Candidate sequences are then delivered for each design profile. Delivered candidates for the same profile must differ by at least a minimum pairwise edit distance. Finally, the generator runs with settings

matched to the task, a species-matched codon table for mRNA and the right length and seed settings for aptamers[31]. The constraint matrix of each campaign is pre-registered, covering for mRNA the GC content, homopolymer runs, stem lengths and codon adaptation of the candidates, and for aptamers their length, GC content, stem lengths, homopolymer runs and batch diversity (Supplementary Methods M2). The generation budget is identical across conditions in every campaign (the budget-matching rule). The campaigns that measure GenNav ran under the pre-registered protocols of Supplementary Methods M2.

Benchmark datasets and anchor sets

All results are obtained on eight benchmark datasets. RNADesign-210, a 210-task benchmark covering the eleven workflows, supports the system-scale evaluation (Fig. 2a), with the full-length tasks under the two evidence-based steps (Supplementary Methods M3). The 31-session process-aware benchmark tests the complete system through the form-driven interface and serves as the task set of the comparison against agent baselines (Fig. 6; Supplementary Methods M4). The constraint-navigation results use a five-protein selection-step pool (Fig. 5a), a 25-target full-length mRNA set (Fig. 5b–d), a 26-target aptamer set (Fig. 5e–g) and an expanded 39-target aptamer pool for the budget sweep (Supplementary Fig. S4). The failure-mode audits and the capability-gating experiments use a 45-task aptamer audit set and a 20-task full-length mRNA audit set (Figs. 3 and 4; Supplementary Methods M1).

External anchor sets provide the reference sequences for capability profiling and for the biological-validity comparisons, and every anchor has a label and a source citation. The CDS-quality comparison uses 20 wild-type CDSs from NCBI RefSeq and 200 random synonymous sequences[32]. Vaccine-mRNA anchors are the assembled BNT162b2 and mRNA-1273 sequences[33]. Published aptamers are the thrombin-binding aptamer, pegaptanib and a SARS-CoV-2 nucleocapsid binder, each with its literature Kd, and the broader groups of aptamer binders and negatives derive from literature affinities and Protein Data Bank (PDB) structures[20,34,35]. The internal ribosome entry site (IRES) anchor set contains nine validated IRES 5′UTRs, six viral and three cellular, against 50 random human 5′UTRs[36–38]. The small-molecule anchor set consists of three binder and non-binder pairs, the theophylline aptamer versus caffeine (PDB 1O15), HIV-1 TAR with aminoglycoside binders (1ANR) and the guanine riboswitch versus adenine and caffeine (1Y27)[39–42]. The mRNA-stability anchor set of 400 sequences with measured degradation rates comes from the stability predictor's public test set, and the measured-MRL 5′UTR libraries come from a published massively parallel translation assay (Gene Expression Omnibus (GEO) accession GSE114002)[26,43].

Statistical analysis

Paired comparisons used two-sided Wilcoxon signed-rank tests with Cliff's δ as the effect size, and multiple-testing control used Benjamini–Hochberg false discovery rate (FDR)[44–46]. Unpaired comparisons used two-sided Mann–Whitney tests[47]. AUROC confidence intervals were bootstrapped 10,000 times, and claim false-positive rates are given Wilson 95% confidence intervals[48,49]. Predictor ranking grades used Spearman correlation against measured values (Kd or MRL)[50]. The GenNav primary comparisons used exact paired McNemar tests on whether each target and profile passed all of its constraints[51]. The baseline comparison (Fig. 6) used the same tests on the delivered validity of each turn ($n = 96$). Thresholds were pre-registered per campaign ($\delta \geq 0.3$ for pilot go gates, AUROC above 0.8, ρ at least 0.6, and FDR $q < 0.05$ with $|\delta|$ at least 0.3 for design gains).

Data availability

All data supporting the findings of this study have been deposited in the Zenodo repository under DOI 10.5281/zenodo.22736194 (CC-BY-4.0). The deposit contains the eight benchmark datasets, the 210-task RNADesign-210 system benchmark, the 31-session process-aware benchmark, the five-protein selection pool, the 25-target full-length mRNA set, the 26-target aptamer set, the 39-target aptamer budget-sweep pool, the 45-task aptamer claim-audit set and the 20-task full-length mRNA claim-audit set. The deposit also contains the external anchor sets (Methods), the packaged capability map with its generator and the per-experiment model-usage register.

Code availability

The complete RNADesignAgent system is available at https://github.com/xlab-BioAI/RNADesignAgent. External research codes are included as pinned submodules, with any local modifications recorded in the deployment guide.

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